Paleomagnetism of late Paleozoic rocks in the northern Cache Creek terrane near Atlin, British Columbia

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The Cache Creek terrane is exposed along the length of the Canadian Cordillera and is composed of oceanic strata that are probably, at least in part, exotic to North America. In the northern portion of the Cache Creek terrane near Atlin, British Columbia, paleomagnetic samples were collected from layered Paleozoic rocks at 22 sites ($\geq$6 samples/site) on Alfred Butte. Principal component analysis of detailed thermal demagnetization data allowed clear isolation of a characteristic remanent magnetization (ChRM) from 17 of these sites. Blocking temperatures to 680°C indicate that this magnetization is carried by hematite, and site-mean ChRM directions are determined with $\alpha_{95} < 10°$ for the majority of sites. On Sentinel Mountain, samples were collected from 16 sites in layered Paleozoic volcanic and chert rocks and from a diabase sill. Thermal demagnetization revealed a ChRM in the chert and volcanic rocks with blocking temperatures to 680°C, whereas alternating-field demagnetization to 40 mT successfully isolated ChRM in the diabase sill. ChRM directions from four sites involved in a mesoscopic S-fold at Alfred Butte fail the fold test, indicating that the ChRM is a postfolding secondary remagnetization. Tests for relative age of structural tilting and remagnetization are ambiguous, with attendant uncertainties in tectonic interpretations. However, rock-magnetic and geologic constraints argue for a chemical remagnetization of these Paleozoic rocks in Late Triassic to Middle Jurassic time, possibly associated with structural juxtaposition of the Cache Creek and Stikine terranes along the Nahlin fault zone. Although certainly nonunique and speculative, the simplest tectonic interpretation of these paleomagnetic data involves postfolding and posttilting remagnetization during the Early Jurassic in a paleolatitudinal position that approximately agrees with predicted North American paleolatitudes for this time.

Le terrane de Cache Creek est exposé le long de l'axe de la Cordillère canadienne, et il est formé de strates océaniques qui, probablement, au moins en partie, ont une origine exotique à l'Amérique du Nord. Dans la région nord du terrane de Cache Creek, près de Atlin en Colombie-Britannique, des échantillons de roches stratifiées paléozoïques ont été prélevés pour étude paléomagnétique à 22 sites ($\geq$6 échantillons/site) sur la butte Alfred. La principale composante qui résulte de l'examen détaillé de la désaimantation par chauffage, indique clairement pour 17 des sites étudiés une direction isolée d'aimantation rémanente caractéristique (ARca). Les températures de blocage à 680°C dévoilent que cette aimantation est portée par l'hématite, et que pour la majorité des sites la moyenne des directions des ARca est déterminée avec $\alpha_{95} < 10°$. Les échantillons du mont Sentinel ont été prélevés à 16 sites, dans des strates paléozoïques de volcanites, de cherts et dans un filon-couche de diabase. La désaimantation par chauffage a mis en évidence dans les volcanites et les cherts une ARca avec des températures de blocage à 680°C, tandis que la désaimantation par champ alternatif à 40 mT a permis d'isoler clairement une ARca dans le filon-couche de diabase. Les directions des ARca de quatre sites de la butte Alfred, dont les roches sont déformées par un pli-S mésoscopique, répondent négativement au test du pli, ce qui indique que l'ARca est une réaimantation secondaire acquise postérieurement au plissement. Les tests appliqués pour définir les âges relatifs du basculement structural et de la réaimantation fournissent des réponses équivoques et créent de nombreuses incertitudes quant aux interprétations tectoniques. Cependant, les propriétés magnétiques des roches et la géologie suggèrent plutôt une réaimantation chimique de ces roches paléozoïques, survenue entre le Trias tardif et le Jurassique moyen, associée probablement à la juxtaposition structurale des terranes de Cache Creek et de Stikine le long de la zone de failles de Nahlin. Quoique spéculative et certainement pas exclusive, l'interprétation tectonique la plus simple de ces données paléomagnétiques implique une réaimantation postérieure au plissement et au basculement durant le Jurassique précoce, en une position géographique paléolatitudinale qui serait, approximativement, d'accord avec les paléolatitudes de l'Amerique du Nord assignées à cette époque.

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Introduction

The Cache Creek terrane is a fault-bounded fragment of upper Paleozoic and lower Mesozoic oceanic strata exposed along the length of the Intermontane Belt in western Canada (Fig. 1). The terrane is characterized by a marine rock sequence and, in places, a Permian Tethyan faunal assemblage. These strata are dissimilar to those in coeval neighboring terranes, as are the Permian faunas (Monger and Ross 1971). The Cache Creek terrane may represent a small portion of ancient ocean floor that travelled from equatorial and (or) western parts of the ancestral Pacific basin and became incorporated into the North American Cordillera in post-Paleozoic time. Alternatively, it may have formed adjacent to the ancient Cordilleran margin as the distal part of a large marginal basin. We attempted to test the "exotic" origin of the Cache Creek terrane by examining paleomagnetic directions recorded in bedded Paleozoic rocks near Atlin, British Columbia.

Along the length of British Columbia, the Cache Creek terrane is made up of three main stratigraphic assemblages (1) mafic volcanic and minor ultramafic rocks; (2) chert and associated pelite; and (3) pure, massive limestone which locally occurs as conspicuous lenses in a matrix of chert or basalt. These strata represent prolonged deposition in an ocean-floor environment, from at least Early Mississippian to Early Triassic time (Monger 1975). Upper Triassic and Lower to Middle Jurassic siliciclastic and volcaniclastic sediments occur locally (Cordey *et al.* 1987) and appear to signify the earliest terrigenous influence into the ancient Cache Creek basin. In southern British Columbia, particularly near the town of Cache Creek, the rocks are highly sheared and characterized as tectonic mélange (Trettin 1980). In northern British Columbia, primary depositional relations are preserved and shearing is confined to discrete fault zones. Mapping and paleomagnetic sampling were conducted in the Atlin Block at

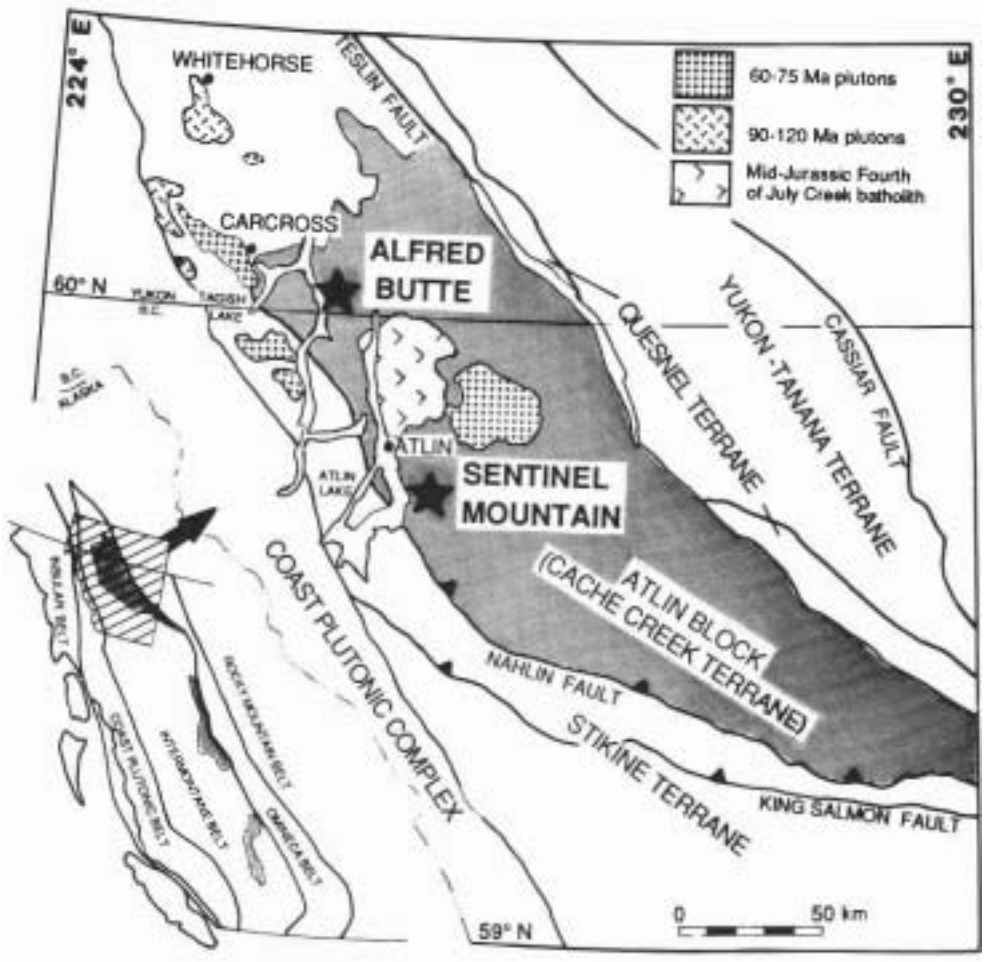


FIG. 1. Location map of Cache Creek terrane (shaded in inset map, lower left) and Atlin Block. Stars locate paleomagnetic sampling areas.

the northern end of the Intermontane Belt. This region provides the most complete and intact geology of the Cache Creek terrane.

The Cache Creek terrane has a much different character from the Stikine and Quesnel terranes, which occur along its western and eastern margins, respectively (Figs. 1, 2). Upper Paleozoic portions of the Stikine and Quesnel terranes are coherent sequences of marine and possibly nonmarine rocks, distinguished by abundant calc-alkaline volcanic and volcaniclastic rocks and limestones with a McCloud-type faunal assemblage (Monger and Berg 1987; Miller and Wright 1987). In contrast, the intervening Cache Creek terrane contains little arc-related material. In many areas, it is structurally disrupted and contains upper Paleozoic fusulinid and coral species that are more akin to coeval faunas found in eastern Asia (Monger and Ross 1971). There is no evidence of a shared history between Cache Creek and adjacent Stikine and Quesnel terranes until Late Triassic time. In southern British Columbia, Upper Triassic volcanic clasts of Quesnel are found in Triassic mélange of the eastem Cache Creek terrane (Shannon 1982). Also, clasts of chert, basalt, limestone, and ultramafic rock probably derived from the Cache Creek terrane are found in Upper Triassic strata of Quesnel (Monger and Berg 1987). In northern British Columbia, a roughly coeval tie may exist between Stikine and Cache Creek greywackes (Monger and Berg 1987; Jackson *et al.* 1991). In addition, Cache Creek detritus is recognized in Middle Jurassic rocks of the Bowser Basin, which lies on Stikine (Monger and Berg 1987), and an Early Jurassic reverse fault juxtaposes northern parts of the Cache Creek and Stikine terranes. Thus it appears that the Cache Creek terrane evolved independently of adjacent terranes until Late Triassic time. Thereafter the Cache Creek terrane was probably amalgamated with Stikine and Quesnel to form Terrane I (Monger *et al.* 1982).

Previous paleomagnetic results

Although no paleomagnetic data are available from the Cache Creek terrane itself, numerous paleomagnetic investigations have been carried out in the Stikine and Quesnel terranes (Fig. 2). Preliminary results from volcanic rocks of the Permian Asitka Group of easternmost Stikine yield paleolatitudes that are roughly concordant with North American paleolatitudes in the Permian (Irving and Monger 1987). Substantial paleomagnetic data are available from Upper Triassic and Lower Jurassic volcanic rocks of Stikine (Takla and Hazelton groups) (Monger and Irving 1980; Vandall and Palmer 1990). The observed paleolatitudes are concordant with expected paleolatitudes determined from paleomagnetism of coeval formations of the Colorado Plateau (Gordon *et al.* 1984; May and Butler 1986). From the southern portion of the Quesnel terrane, paleomagnetic data are available from two Early Jurassic plutons (Copper Mountain and Guichon batholiths) and from bedded volcanic rocks (Nicola Volcanics) interpreted as having been remagnetized during the Early



FIG. 2. Locations of paleomagnetic studies in Terrane I (patterned area). CC, Cache Creek terrane; ST, Stikine terrane; QN, Quesnel terrane.

Jurassic (Symons 1971, 1973*a*, 1983*a*, 1985; Symons and Litalien 1984). Again, the observed paleolatitudes are concordant with expected North American Early Jurassic paleolatitudes (May and Butler 1986; Butler *et* al. 1989). Collectively these paleomagnetic data suggest that Terrane I, including Cache Creek, was near its current latitudinal position with respect to North America by Early Jurassic time.

Additional paleomagnetic data for Terrane I come from the Topley Intrusions, which occur along the Stikine – Cache Creek contact in central British Columbia (Symons 1973*b*, 1983*b*). However, the significance of these data is uncertain. The ages of crystallization and magnetization of the intrusions are poorly defined, with K–Ar dates ranging from 112 to 163 Ma, and the paleohorizontal at the time of magnetization is unconstrained.

Paleomagnetic data have been obtained from the Upper Cretaceous Carmacks Group (Marquis and Globerman 1988), which overlaps the northern Cache Creek and Stikine terranes in the Whitehorse Trough (Fig. 2). These volcanic rocks are layered and radiometrically dated at 70.4 ± 2.4 Ma. Marquis and Globerman (1988) interpreted the results as indicating 13.4 ± 8.5° of northwards transport relative to the craton since 70 Ma. However, this interpretation is not without controversy because it depends on the statistical analysis applied to the paleomagnetic data from the Carmacks Group and on the 70 Ma North American reference paleomagnetic pole used in the analysis (Butler 1990; Marquis *et* al. 1990). When compared with a recently published paleomagnetic reference pole from Upper Cretaceous volcanic rocks in Montana (Gunderson and Sheriff 1991), the paleomagnetic pole from the Carmacks Group indicates 9.8 ± 9.9" (95% confidence limits) of apparent northward transport. Although the paleomagnetic pole from the Carmacks Group is concordant with this Upper Cretaceous reference pole in a formal statistical sense, ~1000 km of northward transport after 70 Ma is suggested.

Geology and sampling

The Atlin Block is bounded on the west by the northeast-dipping Nahlin Fault and on the east by the subvertical Teslin Fault (Fig. 1). The rocks of the Atlin Block are divided into three main facies belts (Monger 1975). The southwestern facies belt is characterized by Carboniferous basalt, metabasalt, and basaltic flow breccia (Nakina Formation); bedded chert with interbedded pelite (Kedahda Formation); and ~1500 m of pure, massive limestone of Late Mississippian to Permian age (Horsefeed Formation). The central facies belt consists mainly of ribbon chert that is locally interbedded with volcanic greywacke and includes lenses of Mississippian and Permian limestone. The northeastern facies belt contains Permian chert, pelite, and subordinate interbedded greywacke (Kedahda Formation), which grades upwards and laterally into mid- and Upper Permian carbonate rocks of the Teslin Formation. Paleomagnetic samples were collected on Alfred Butte, which lies in the southwestern facies belt, and on Sentinel Mountain, which straddles the transition between the central and southwestern facies belts (Fig. 1).

Alfred Butte

Alfred Butte lies 50 km north-northwest of Atlin, British Columbia, immediately north of the British Columbia – Yukon border (Fig. 1). The area was mapped at a scale of 1 : 12 500 to provide an accurate geologic framework for paleomagnetic sampling (Fig. 3). Alfred Butte is a north- to northwest-dipping monocline composed of a thick section of pure, massive upper Paleozoic limestone (Horsefeed Formation) and capped by a thin but conspicuous series of reddish tuffs, flows, and interlayered tuffaceous limestone. A west-vergent thrust fault occurs near the top of the volcanogenic section. This fault has evidently been modified by a younger tilting event that produced the dominant monoclinal structure of the butte. Narrow diorite dikes intrude the section and, in turn, are crosscut by a series of north-striking normal faults, each of which accommodates a few metres of offset. Paleomagnetic samples were collected in the volcanogenic rocks at the southwest end of Alfred Butte where they are well exposed and relatively undisturbed by the normal faults and diorite dikes.

Twenty-two paleomagnetic sites (minimum 6 samples/site) were collected across approximately 50 m of stratigraphic section on Alfred Butte (Fig. 3). Core samples of 2.5 cm diameter were drilled in *situ* with standard paleomagnetic coring equipment, and samples were oriented with a magnetic compass and inclinometer. The sampled volcanic rocks occur as three discrete tongues in a sequence of fusulinid-rich, well-bedded to massive gray limestone. These three horizons were subdivided into eight individual igneous cooling units, based on metre-scale textural and compositional divisions. In addition to the igneous rocks, intervening limestone layers were also sampled, particularly toward the top of the volcanogenic section where the carbonate is enriched in visible hematite concentrated along stylolites.

The depositional age of the volcanogenic section is well constrained by diagnostic Late Pennsylvanian fusulinids (Monger 1975) and by three conodont samples collected for this study and identified by M. J. Orchard of the Geological Survey of Canada. Two conodont samples from near the top of the vol-

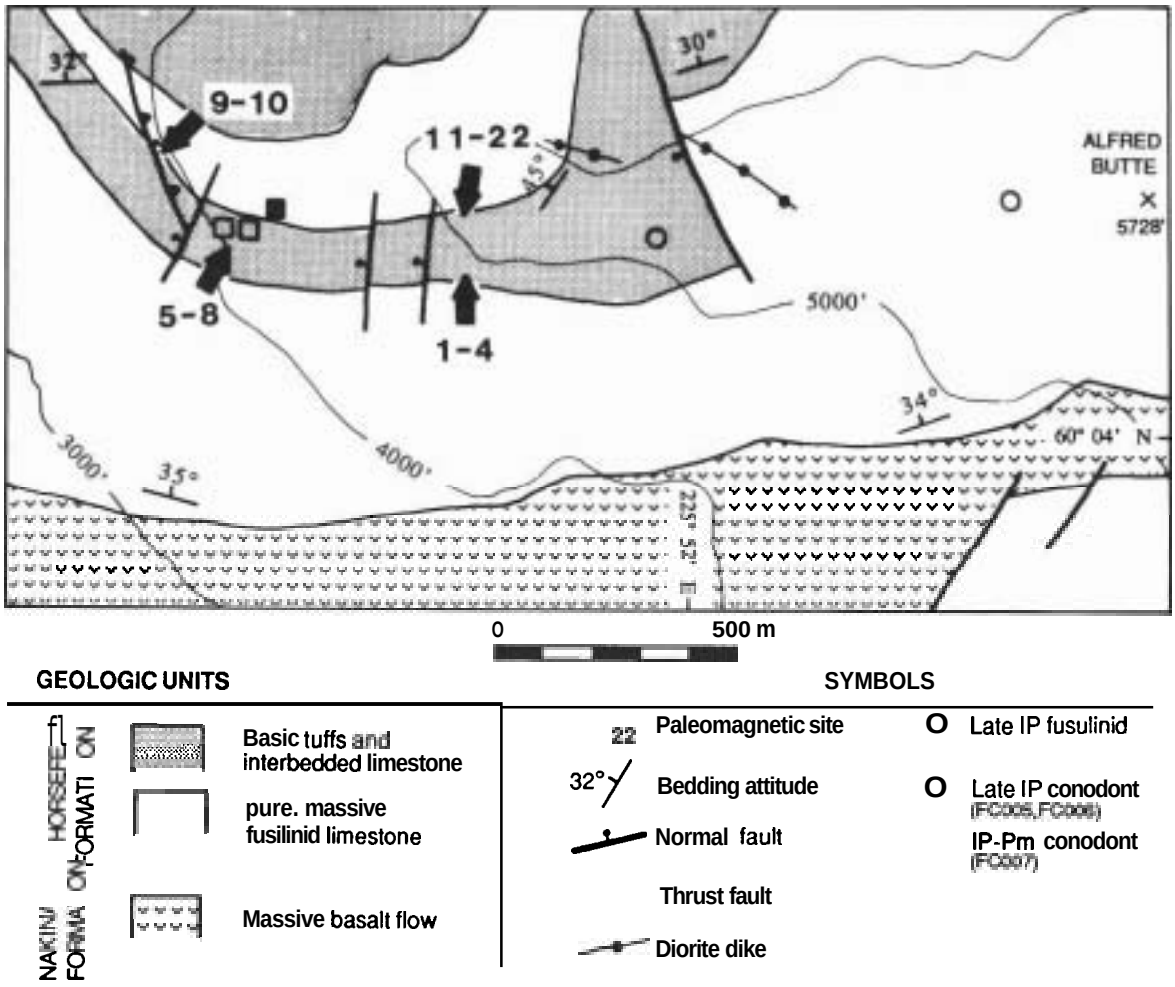


FIG. 3. Geology and sampling locations at Alfred Butte. Bold numbers indicate paleomagnetic sampling site designation (e.g., 1 = site AB001; see Table 1). Fossil numbers in parentheses (e.g., FC007) correspond to Geological Survey of Canada locality numbers.

canogenic section (Geological Survey of Canada samples FC005 (= C-156727) and FC006 (= C-156728)) contained species that are Middle to Late Pennsylvanian. A third sample (Geological Survey of Canada sample FC007 (= C-156729)) from immediately above the volcanogenic section also contained Middle to Late Pennsylvanian conodont species. Well-bedded limestone and tuff layers provide original horizontal control for individual paleomagnetic sites. Graded bedding in the tuffs indicates that the section is right side up.

The sampled volcanic rocks are basaltic in composition (see Table 1 of Monger 1975). In thin section, the flow rocks are porphyritic with saussuritized feldspar pseudomorphs in an aphanitic matrix now composed of fine mica and calcite. The pyroclastic rocks contain altered lithic fragments, largely identical to underlying flow lithologies, in a groundmass of fine mica and calcite. Although primary textures are preserved, the limestones are partially recrystallized and the tuffs and flows contain visible interstitial and intragranular hematite.

Sentinel Mountain

Sentinel Mountain lies 15 km south of Atlin, British Columbia, on the east side of Atlin Lake (Fig. 1). This area was mapped at a scale of 1 : 12 500 to provide a geologic base map for paleomagnetic sampling (Fig. 4). The southern part of the area is occupied by a highly sheared and chloritized assemblage of basaltic flow breccias that enclose resistant olistostromes of brecciated chert and recrystallized limestone. To the north, riding in fault contact above these penetratively sheared volcanic rocks, is an apparently younger assemblage of bedded cherts, pillow basalt, and diabase sills. Paleomagnetic sampling was conducted in the upper plate rocks, because they retain a comparatively intact stratigraphy.

The layered volcanic rocks and chert of the upper plate are intruded by broad sills of diabase at regular intervals. Sixteen paleomagnetic sites (minimum 6 samples/site) were collected on Sentinel Mountain within and along the margins of a 50 m wide, subvertical diabase sill that intrudes radiolarian chert and volcanic rocks (Fig. 4). Along its upper contact, compositional layering in the sill dips steeply northward and is coplanar with bedding in the adjacent chert. Along its lower contact, the sill follows a thin zone of chert and graded volcanic tuff. The upper and lower contact rocks are visibly baked within 10 m of the diabase body. The baked zone is rich in hematite, some of which apparently formed as a result of sill emplacement.

Although the exact age of the diabase is unknown, field relations suggest that it is only slightly younger than the submarine rocks that it intrudes. Unlike younger batholiths and dikes that crosscut stratigraphic and structural trends in the Cache Creek Group, the diabase sills strictly follow bedding in the

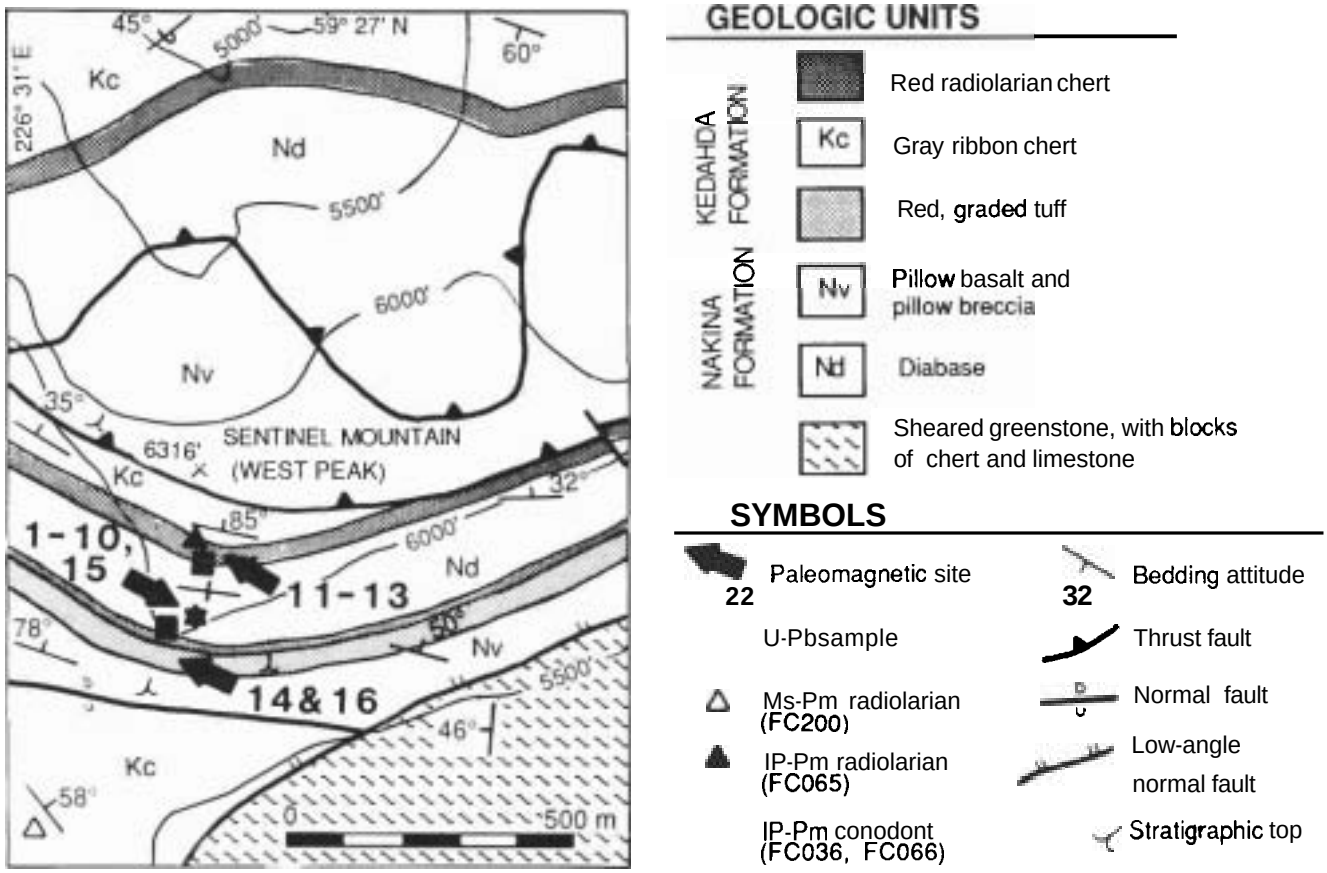


FIG. 4. Geology and sampling locations at Sentinel Mountain. Bold numbers indicate paleomagnetic sampling site designation (e.g., 1 = site ST001; see Table 2). Fossil numbers (except FC200) correspond to Geological Survey of Canada locality numbers.

Paleozoic chert and basalt. These sills do not cut the faults that bound the chert and basalt sequence but instead terminate at those structures. Most importantly, the diabase is laterally transitional into layered basalt flows that are clearly depositional in the Paleozoic sequence. The diabase is slightly coarser, but compositionally equivalent to the layered basalt and apparently was derived from the same magma. These relationships indicate a temporal link between the diabase sills and the layered Paleozoic rocks.

The depositional age of the chert and volcanic rocks was previously unknown, but conodonts and radiolarians collected in the course of this study have been identified as Permian and possibly Late Pennsylvanian (Fig. 4). Two red chert samples, one near the base (Geological Survey of Canada sample FC036 (= C-156733)) and the other near the top (Geological Survey of Canada sample FC066 (= C-156737)) of the diabase sill, yield probable Pennsylvanian and Permian conodonts, respectively (M. J. Orchard, persona1 communication, 1991). A third red chert sample from Geological Survey of Canada locality FC065, also from near the top of the sill, produced diagnostic Late Pennsylvanian – Early Permian radiolarians (sp. *Pseudoalbaillella*; T. A. Harms, written communication, 1990). Two additional red chert samples (FC054, FC055 (= C-156735)) were collected adjacent to the diabase sill on the east peak of Sentinel Mountain, 2 km east of the paleomagnetic sampling area. These produced Late Pennsylvanian – Early Permian radiolarians and Permian conodonts, respectively. Our inference is that the diabase is also Late Pennsylvanian – Early Permian based on the field relations described above. It appears that the diabase sills on Sentinel Mountain always occur at the same biostratigraphic level and may represent a single body repeated by numerous thrust faults (see also Bloodgood et al. 1990).

Paleomagnetism

All paleomagnetic specimens were stored, measured, and demagnetized in a magnetically shielded room with average field intensity less than 200 nT. All measurements of natural remanent magnetization (NRM) were done using a two-axis cryogenic magnetometer (ScT C-102). Following initial NRM measurements, pilot specimens from all sites were subjected to progressive thermaland alternating-field (AF) demagnetization to determine blocking temperature and coercive force spectra for the NRM. Progressive thermal demagnetization employed 14 temperature steps between 100 and 680°C. Magnetic field inside the furnace was less than 5 nT. Progressive AF demagnetization was conducted using a tumbling-specimen demagnetizer (Schonstedt GSD-5) in a series of 12 peak-field steps between 1.2 and 90 mT. For most of the Atlin lithologies, thermal demagnetization was the more effective method for isolating the characteristic remanent magnetization (ChRM); the diabase was the single lithology that responded better to AF demagnetization.

Demagnetization behavior

Five sites were drilled in limestone lacking visible hematite pigment near the base of the volcanogenic section at Alfred Butte. Specimens from these sites had very low NRM intensity

($< 10^{-4}$ A/m), which did not allow progressive demagnetization and are not considered further. The remaining samples from Alfred Butte (7 sites in limestone containing visible hematite pigment and 10 sites in volcanic rocks) had NRM intensities sufficient for progressive demagnetization.

The NRM of pilot specimens of the volcanic rocks and hematitic limestones from Alfred Butte were virtually unaffected by AF demagnetization to peak fields of 90 mT. The same specimens displayed a steady decay in NRM intensity and good directional stability during thermal demagnetization (Fig. 5). Typical behavior for the volcanic rocks is illustrated in Figure 5*a*. A ChRM is quite clearly isolated by thermal demagnetization to 600°C and a significant portion of the NRM has blocking temperatures between 600 and 680°C. Typical behavior for the tuffaceous limestones is illustrated in Fig. 5b. Note the slight changes in direction for demagnetization temperatures below −530°C and excellent directional stability from 560 to 670°C. Some of the limestone specimens had a component of viscous magnetization acquired parallel to the geomagnetic field at the sampling site. This component was removed by demagnetization to 500°C. These observations indicate that hematite is the carrier of the ChRM.

For sites from Alfred Butte, all samples were demagnetized at multiple temperatures within the blocking temperature spectrum of the ChRM. Principal component analysis (Kirschvink 1980) was used to determine least-squares line fits to the progressive demagnetization data. For most specimens, ChRM measurements from three to four temperature steps were used for principal component analysis. The origin option (which treats the origin of the vector component diagram as an equally weighted data point) was used in these analyses. The resulting sample ChRM directions were then used to determine site-mean ChRM directions and virtual geomagnetic poles (VGP's) using Fisher statistics (Table 1). Some sites contained samples with ChRM directions that deviated by more than two estimated angular standard deviations from the site-mean direction. These outliers were excluded from site-mean calculations. The ChRM directions are generally well grouped within each site, with $\alpha_{95} < 10°$ for the majority of sites. Two sites (AB001 and AB004) had aberrant site-mean ChRM directions compared with other sites from Alfred Butte. These sites have been excluded from further calculations.

Eleven sites from Sentinel Mountain were collected through the 50 m width of the diabase sill; three sites were collected in red radiolarian chert near the upper contact of the sill; and the remaining two sites were collected in baked volcanic rocks near the lower contact of the sill. Many diabase specimens displayed unstable directions of NRM when heated to temperatures in excess of 500°C. However, AF demagnetization to 40 mT peak field successfully isolated a ChRM from the majority of samples from the diabase sill (Fig. 6*a*). Several diabase sarnples exhibited unstable NRM directions during AF demagnetization and results from these samples were excluded from determination of site-mean ChRM directions. The demagnetization behavior of diabase samples indicates that the ChRM is carried by magnetite. Each specimen from the diabase sill was demagnetized in multiple AF steps, and principal component analysis of results from four to seven AF steps was used to determine the specimen ChRM direction.

Chert and volcanic rocks from along the contacts of the diabase sill on Sentinel Mountain were treated with thermal demagnetization, and typical behavior is shown in Fig. 6b. A

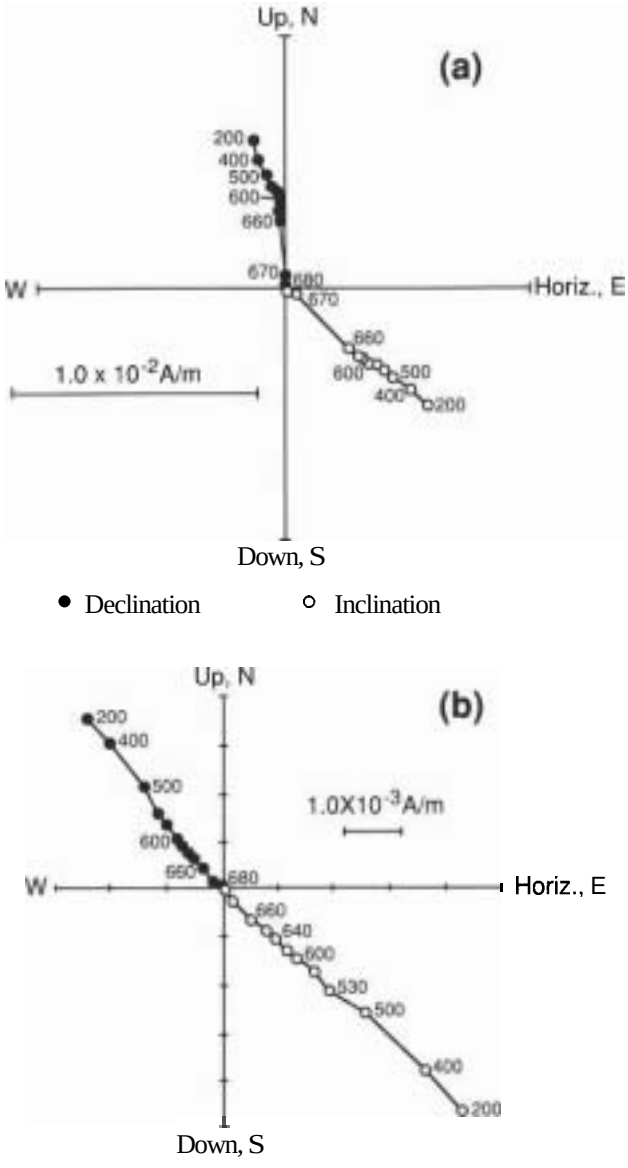


FIG. 5. Vector component diagrams of demagnetization behavior for Alfred Butte volcanic tuff (a) and hematitic limestone (b). ●, projections onto the horizontal plane; ○, projections onto the vertical plane. Numbers adjacent to data points indicate temperature of thermal demagnetization in °C.

ChRM was isolated at temperatures ≥600°C and maximum blocking temperatures are ≤680°C, indicating that hematite is the carrier of the ChRM. All samples were demagnetized at multiple temperatures within the blocking temperature spectrum of the ChRM. Principal component analysis of ChRM at four to five temperature steps was used to determine specimen ChRM directions. Resulting site-mean ChRM directions and VGP's for sites from Sentinel Mountain are listed in Table 2.

Local fold *test*

Sites AB006, AB007, AB009, and AB010 were collected from the limbs of a mesoscopic S-fold on the far west end of Alfred Butte. The fold is associated with a west-vergent thrust fault and involves several metres of volcanogenic strata. Based on this geometry, the fold appears to predate northward tilting of Alfred Butte. When the axes of the fold are corrected for the homoclinal tilting of Alfred Butte, they restore to horizon-

TABLE 1. Site-mean directions and virtual geomagnetic poles for Alfred Butte

Site no.	Lith.	N	R	α_{95} (°)	J ($\times 10^{-4}$ A/m)	I (°)	D (°)	Lat. (°N)	Long. (°E)	I' (°)	D' (°)	Lat.' (°N)	Long.' (°E)
AB001[a]	volc	4	3.9752	9.2	112.5	29	290	23	126	0	295	12	113
AB002	volc	4	3.9561	11.2	0.7	37	330	45	88	0	335	27	74
AB003	volc	7	6.9772	3.7	23.1	50	337	57	84	12	344	34	66
AB004[a]	volc	6	5.8588	12.7	39.1	−31	46	5	3	−27	67	−1	343
AB005	volc	6	5.9765	4.6	41.8	43	2	55	43	13	5	36	40
AB006	volc	4	3.9885	5.7	32.8	44	353	55	57	11	347	35	62
AB007	volc	4	3.9301	14.2	30.7	52	351	62	62	21	4	41	41
AB008	volc	5	4.9247	10.6	35.8	45	318	46	106	27	332	40	83
AB009	volc	6	5.9697	5.2	19.0	46	352	57	58	13	346	35	64
AB010	volc	6	5.8485	14.2	26.9	39	350	52	60	−14	336	20	71
AB016	1st	8	7.9830	2.7	6.0	70	355	83	70	34	337	46	78
AB017	1st	8	7.9377	5.3	16.8	66	351	77	72	29	334	42	80
AB018	1st	8	7.9288	5.6	10.5	49	324	51	100	0	325	24	85
AB019	1st	8	7.9435	5.0	6.0	45	328	50	93	−4	327	23	82
AB020	1st	7	6.9157	7.2	9.3	49	328	53	96	0	327	25	83
AB021	1st	8	7.9323	5.5	5.7	41	324	46	97	−8	324	20	85
AB022	1st	8	7.9262	5.7	4.6	44	324	48	98	−5	324	21	85

NOTES: Lith. is lithology of sampled horizon (Ist, limestone; volc, volcanic rock); N is number of specimens at each site; R is resultant vector length of the N unit vectors; α_{95} is the radius of cone of 95% confidence; J is site-mean intensity over the interval of demagnetization steps used for principal component analysis; I and D are inclination and declination of the site-mean direction; Lat. and Long. are latitude and longitude of virtual geomagnetic pole; 1' and D' are inclination and declination corrected for bedding tilt; Lat.' and Long.' are latitude and longitude of virtual geomagnetic pole determined from tilt-corrected site-mean direction.

[a]Site rejected from calculation of formation mean direction and paleomagnetic pole.

tal attitudes and become parallel to dominant structures in the region. A complete structural correction for the sites involved in the fold requires at least a two-stage structural correction: (1) untilting of the broad-scale monoclinal tilt that has affected the entire section (thereby restoring the fold axis to horizontal); then (2) unfolding to restore the limbs to horizontal.

Figure 7 illustrates the results of this fold test. The *in situ* site-mean directions of the four sites distributed through the fold are well grouped; the best estimate of Fisher's precision parameter (k) for the four *in situ* site-mean directions is 215.8. The structurally corrected site-mean directions are quite scattered, with k = 18.9. These observations constitute a dramatic failure of the fold test of paleomagnetic stability. The site-mean directions fail the fold test at greater than 99% confidence according to either the formulation of Watson and Irving (1957) as tabulated by McElhinny (1964) or the formulation of McFadden and Jones (1981). The implication is that the ChRM for the rocks involved in the fold was acquired after the folding; this ChRM is clearly a postfolding secondary remagnetization.

Because the *in situ* site-mean directions and NRM characteristics are quite consistent throughout the Alfred Butte section, we infer that the ChRM of the entire section is a secondary remagnetization. As described below, the *in situ* ChRM directions from the sites at Sentine1 Mountain are in broad agreement with those from Alfred Butte. Coupled with the observation that all site-mean ChRM directions from both collecting locations are of normal polarity, we further conclude that the ChRM obsemed at both locations is of secondary rather than primary origin. This means that we cannot determine the original Paleozoic paleolatitude of the Cache Creek terrane because apparently no primary magnetization has survived. Furthermore, we have no direct control on either the age of magnetization or the paleohorizontal at the time of remagnetization. Consequently further analysis and interpretations of these paleomagnetic data require indirect inferences, with resulting uncertainties.

Regional tilt test

Figure 8 illustrates the *in situ* site-mean ChRM directions observed for sites from Alfred Butte and Sentine1 Mountain. Mean *in situ* directions computed from each of these two regions are also shown in Fig. 8 and are listed in Table 3. These mean directions agree closely in inclination but have distinct declinations. The angle between the two mean directions is 12.2" and the mean directions are significantly different at the 99% confidence level (McFadden and Lowes 1981). We interpret the agreement in inclination of *in situ* ChRM between the two sampling areas as indicating a common time and mode of remagnetization. The age of remagnetization is likely similar, especially considering that all sites from both areas have normal polarity. In addition, the paleohorizontal at the time of remagnetization of these two areas was similar as discussed below. The observed difference in the mean *in situ* declinations between the Alfred Butte and Sentinel Mountain data sets may be the result of differential vertical-axis rotation between the two areas. Such vertical-axis tectonic rotations are often observed in areas affected by strike-slip faulting (Irving 1979; Beck 1980).

The Alfred Butte section dips northwest an average of 35° about an average strike of N60°E and the Sentine1 Mountain section dips north at ~70° about a strike of N87°W. This difference in mean bedding attitude suggests that a regional tilt test might allow the relative timing of ChRM formation and differential tilting between the two areas to be addressed. Was the ChRM acquired while the beds were in their current *in situ* attitudes? Or did the ChRM acquisition predate the differential tilting between the two areas? From the local fold test described above, we know that ChRM acquisition postdates development of the fold at the west end of Alfred Butte. So in performing the regional tilt test, ChRM directions from the sites involved in the fold must be corrected for the average tilt of Alfred Butte (while the fold limbs are left intact). We have performed the regional tilt test using a number of methods. No matter how the analysis is approached, the results are ambigu-

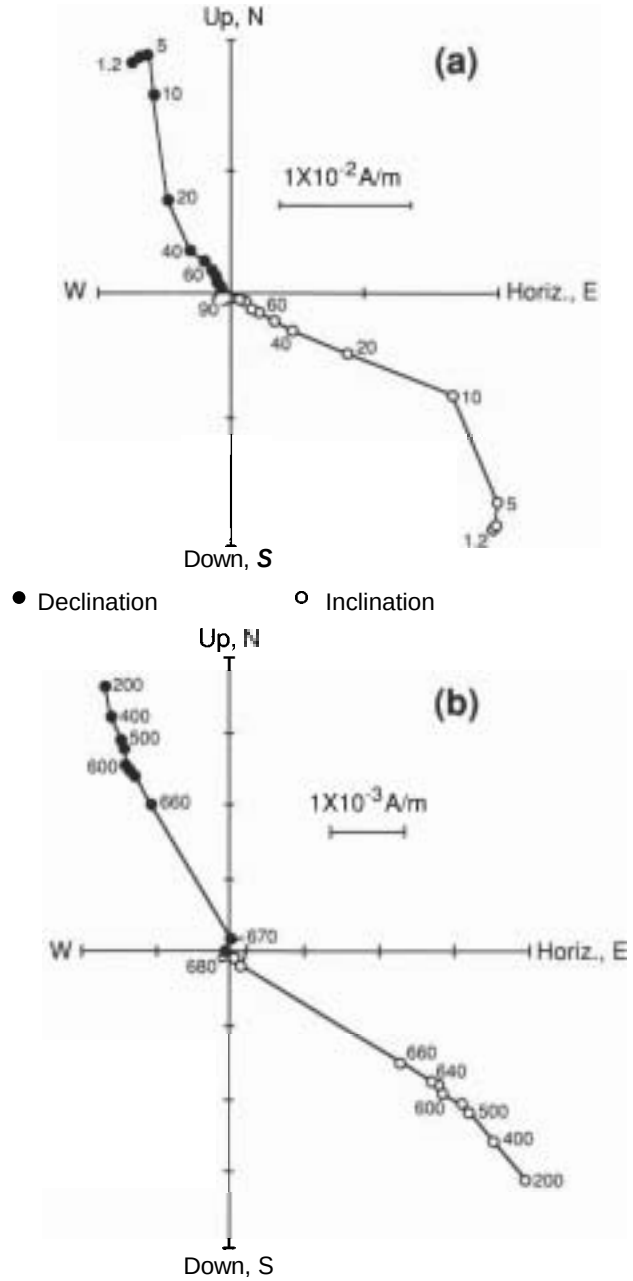


FIG. 6. Vector component diagrams of demagnetization behavior for Sentine1 Mountain diabase *(a)* and chert ***(b)***. Symbols as in Fig. 5. Numbers adjacent to data points in ***(a)*** indicate peak AF in mT; numbers adjacent to data points in (b) indicate temperature of thermal demagnetization in °C.

ous and do not provide a robust result. We recount only the most fundamental results here.

With the present data, the simplest regional tilt test is to compare the clustering of site-mean ChRM directions before and after tilt correcting the bedding at each site to horizontal (while leaving the Alfred Butte fold intact). As listed in Table 3, the best estimate of Fisher's precision parameter, k, for the 31 ***in situ*** site-mean ChRM directions is 28.9, whereas k = 18.9 following tilt correction. The observed ***decrease*** in k resulting from tilt correction is significant at the 5% level but not at the 1% level. This result indicates failure of the regional tilt test at 95% confidence level, implying that the ChRM was acquired ***after*** differential tilting of the two areas. However, repeating the analysis using site-mean VGP's leads to the opposite conclusion. As shown in Table 3, the best estimate of Fisher's precision parameter, K, for the 31 ***in situ*** VGP's is 19.9, whereas K = 31.7 following tilt correction. The observed ***increase*** in K resulting from tilt correction is significant at the 5% level but not at the 1% level. The result of analyzing the VGP dispersion indicates passage of the regional tilt test at 95% confidence level, implying that the ChRM was acquired ***prior*** to differential tilting of the two areas.

Comparisons of ***in situ*** and tilt-corrected mean directions and paleomagnetic poles from Alfred Butte and Sentine1 Mountain (Table 3) do not provide a conclusive regional tilt test. Applications of Bingham statistics (Onstott 1980) and other approaches do not clarify the results. Also, we are not able to effectively address the possibility of syndeformational origin of the ChRM (e.g., Irving and Thorkelson 1990) because there is insufficient variation in structural attitude both within and between the two areas. The ambiguous results of the regional tilt test mean that we are unable to determine whether remagnetization occurred before, during, or after the sampling areas were tilted. Remagnetization could have occurred any time after development of the S-fold on Alfred Butte. In the discussion below, we address indirect evidences for the origin and timing of magnetization.

Paleomagnetic poles

By averaging site-mean VGP's from each of the two sampling areas and from the combined data sets, paleomagnetic poles were determined from the ***in situ*** and the tilt-corrected data sets. These paleomagnetic poles are listed in Table 3 along with statistical parameters and are illustrated in Fig. 9*a*. Paleomagnetic poles determined from the combined data sets are compared with Mesozoic North American paleomagnetic poles from the continental interior in Fig. 9*b*. Comparisons between the North American paleomagnetic poles and the paleomagnetic poles from the Atlin region of the Cache Creek terrane are an important element of interpretations discussed below.

Discussion

Age and origin of remagnetization

To provide a regional context for discussion of the age and origin of remagnetization of the Cache Creek terrane in the Atlin region, we describe the major structural and thermal events that have affected this area. One of the most important structures is the Nahlin Fault (Wheeler 1961), which juxtaposes the Cache Creek and Stikine terranes (Monger ***et al.*** 1978; Bultman 1979). Compressional deformation in the Nahlin Fault zone involves strata as old as Late Triassic (Souther 1971; Monger 1975; Bultman 1979) and as young as Middle Jurassic (Takwahoni Formation). The S-fold on Alfred Butte appears kinematically coordinated with nearby folds in the Nahlin Fault zone, and we therefore interpret it as a Late Triassic to Middle Jurassic structure.

The age of tilting is less certain because no untilted rocks occur at either Sentine1 Mountain or Alfred Butte. Tilting may have occurred during the late stages of activity on the Nahlin Fault zone, or it may be much younger. Because most Upper Cretaceous, Paleocene, and Eocene strata in this region are subhorizontal (Aitken 1959), we infer that tilting of Alfred Butte and Sentine1 Mountain occurred prior to latest Cretaceous time.

The age of remagnetization of the Cache Creek rocks in the Atlin area is constrained as younger than the time of develop-

TABLE 2. Site-mean directions and virtual geomagnetic poles for Sentinel Mountain

Site no.	Lith.	N	R	α_{95} (°)	I ($\times 10^{-3}$ A/m)	I (°)	D (°)	Lat. (°N)	Long. (°E)	I' (°)	D' (°)	Lat.' (°N)	Long.' (°E)
ST001	diab	8	7.9003	7.8	55.2	38	311	39	110	−12	323	18	85
ST002	diab	7	6.7692	14.3	10.9	38	328	46	92	−20	334	17	73
ST003	diab	8	7.9691	4.4	21.9	38	322	43	98	−18	330	17	77
ST004	diab	8	7.9713	4.2	73.8	38	307	37	115	−9	322	19	87
ST005	diab	5	4.9444	12.5	18.0	52	308	47	122	−1	333	27	77
ST006	diab	7	6.9148	9.1	10.3	44	305	40	120	−4	325	23	85
ST007	diab	5	4.9226	13.1	5.7	41	285	27	137	7	315	24	97
ST008	diab	5	4.9142	17.2	2.3	60	312	55	127	3	340	30	70
ST009	diab	4	3.9811	10.4	8.8	56	309	50	124	1	336	28	74
ST010	diab	6	5.9821	5.4	14.1	51	345	61	73	−14	352	23	56
ST011	cht	5	4.9119	14.9	29.6	47	324	51	100	0	338	28	72
ST012	cht	6	5.9815	5.4	54.0	49	336	56	86	−18	343	20	64
ST013	cht	6	5.9500	10.7	36.7	64	316	61	129	−5	344	27	64
ST014	cht	6	5.8943	16.3	16.9	67	343	77	99	2	357	32	51
ST015	diab	8	7.8878	9.2	37.1	48	326	52	98	−12	339	22	70
ST016	volc	6	5.9099	13.2	25.1	57	352	67	63	−7	0	27	46

NOTES: diab, diabase sill; **cht, chert; see Table 1 notes for further explanations.**

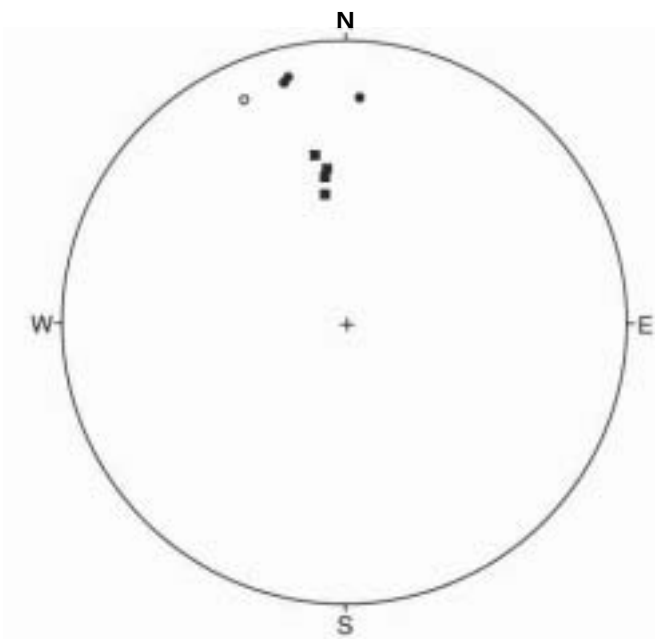


FIG. 7. Fold test results from Alfred Butte. Equal-area projection shows site-mean directions ***in situ*** (■) and following tectonic correction (○, ●). Filled symbols indicate directions in the lower hemisphere; open symbol indicates direction in the upper hemisphere.

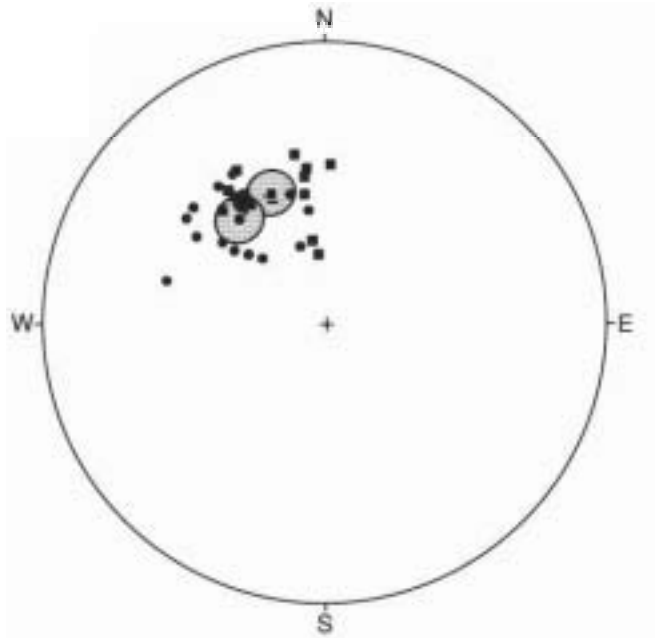


FIG. 8. Equal-area projection shows ***in situ*** site-mean directions from Alfred Butte (■) and from Sentine1 Mountain (●). Mean directions for the two sampling areas are shown surrounded by 95% confidence limits. All directions are in the lower hemisphere.

ment of the S-fold at Alfred Butte. We argued above that this structure is probably Late Triassic to Middle Jurassic in age. Numerous subsequent thermal events have affected the region. However, the remagnetization at Sentine1 Mountain and Alfred Butte probably occurred as a low-temperature chemical remagnetization rather than a high-temperature thermal remagnetization for the following reasons.

Blocking temperatures of the ChRM approach the Curie temperature of hematite (680°C) and thermoviscous remagnetization would require temperatures in excess of 650°C (Pullaiah ***et al.*** 1975). Paleotemperature estimates from metamorphic facies and conodont alteration indices (CAI) indicate that the rocks at Sentine1 Mountain and Alfred Butte have not been heated above ~550°C. Diabase and basalt from Sentine1 Mountain contain minera1 assemblages typical of prehnite–pumpellyite or low greenschist facies (albite, chlorite, clinozoisite, pumpellyite, and sphene), corresponding roughly to temperatures in the 200–400°C range (Turner and Verhoogen 1960). Conodonts from Sentine1 Mountain and Alfred Butte have CAI values between 5 and 6 (M. J. Orchard, persona1 communication, 1989), which indicate that the sampling areas have been heated to >300°C but not more than ~550°C (Rejebian ***et al.*** 1987).

The alternative to thermoviscous remagnetization is that the Atlin rocks have experienced a chemical remagnetization. The hematite carrying the secondary ChRM throughout the Alfred

TABLE 3. Mean directions and paleomagnetic poles from the Atlin block

Data set	N	Mean direction k	α_{95} (°)	I (°)	D (°)	Paleomagnetic pole K	A_m (°)	Lat. (°N)	Long. (°E)
Alfred Butte *in situ*	15	37.0	6.4	48	338	26.8	7.5	57	81
Alfred Butte corrected	15	17.5	9.4	9	338	28.8	7.3	32	73
Sentinel Mountain *in situ*	16	30.6	6.8	51	320	22.0	8.0	52	109
Sentinel Mountain corrected	16	29.5	6.9	−7	337	38.0	6.1	24	72
Combined in *situ*	31	28.9	4.9	50	329	19.9	5.9	55	96
Combined corrected	31	18.9	6.1	1	337	31.7	4.7	28	72

NOTES: N is number of sites in each data set; k is best estimate of Fisher's precision parameter; α_{95} is the radius of cone of 95% confidence; I and D are mean inclination and mean declination; K is best estimate of Fisher's precision parameter for the set of virtual geomagnetic poles (VGP's); A_m is the radius of cone of 95% confidence about the paleomagnetic pole (= mean of the set of VGP's); Lat. and Long. are latitude and longitude of the paleomagnetic pole.

Butte section could be a postfolding chemical precipitate. However, a chemical remagnetization of the Sentinel Mountain section presents a more complex scenario. The ChRM of the Sentinel Mountain cherts is carried by hematite, whereas the ChRM of the Sentinel Mountain diabase is carried by magnetite. A chemical origin of the remagnetization thus requires chemical precipitation of hematite in most lithologies and authigenic magnetite in the diabase. Although the geochemistry is not entirely understood, coeval chemical remagnetization by precipitation of hematite in some rock types and of magnetite in other rock types is documented for late Paleozoic remagnetization during the Alleghenian orogeny (McCabe and Elmore *1989).*

Possible tectonic interpretations

With the ambiguity of the paleohorizontal at the time of remagnetization, there is much uncertainty connected with tectonic interpretation of the paleomagnetic data from the Atlin block. We here address implications of the most likely ages and tectonic settings of remagnetization.

The most conservative interpretation involves Early Jurassic remagnetization of the Atlin Block following folding and tilting. If tilting in the region occurred as part of the juxtaposition of the Cache Creek and Stikine terranes, then Early Jurassic chemical rernagnetization could have affected these rocks in essentially their present structural attitudes. The proximity of the *in situ* paleomagnetic pole from the Atlin Block to the Early Jurassic portion of the North American apparent polar wander path is consistent with this interpretation. For example, the *in situ* paleomagnetic pole from the Atlin Block yields a concordant paleolatitude when compared with the paleomagnetic pole from the Pliensbachian Kayenta Formation (Bazard and Butler *1991).* In this interpretation, Early Jurassic remagnetization could have resulted from chemical reactions involving orogenic fluids flushed through the rocks by thrusting associated with the suturing of the Cache Creek and Stikine terranes. This scenario is similar to the late Paleozoic remagnetization that accompanied thrust faulting and folding during the Alleghenian orogeny (Oliver *1986;* Miller and Kent *1988;* McCabe and Elmore *1989).* It is also possible that intrusion of the Middle Jurassic Fourth of July Creek batholith (Fig. *1;* Mihalynuk *et al. 1991)* induced hydrothermal activity resulting in chemical remagnetization of the Atlin Block.

An Early Jurassic rernagnetization of the Atlin Block in a paleolatitudinal band concordant with expected Early Jurassic paleolatitudes of adjacent North America is consistent with available paleomagnetic data from other regions of Terrane I. Extensive paleomagnetic investigations of the Upper Triassic and Lower Jurassic Takla and Hazelton groups of the Stikine terrane yield concordant paleolatitudes (Monger and Irving *1980;* Gordon *et al. 1984;* May and Butler *1986;* Vandall and Palmer *1990).* The positive fold tests reported by Vandall and Palmer *(1990)* strongly indicate that these volcanic rocks retain a primary magnetization acquired during original cooling. Therefore, in contrast with the situation for the Atlin remagnetization, the paleohorizontal at the time of magnetization is quite securely determined for the Upper Triassic and Lower Jurassic volcanic rocks of the Stikine terrane.

Because the regional tilt test is ambiguous, another possible interpretation of the paleomagnetic data from the Atlin Block is that the remagnetization occurred after folding but before tilting. In such an interpretation, the "tilt-corrected" paleomagnetic pole from the Atlin Block must be compared with Late Triassic and younger North American reference paleomagnetic poles to draw conclusions about tectonic motions following remagnetization (Fig. *9b).* As shown in Fig. *9b,* the North American paleomagnetic poles closest to the corrected Atlin paleomagnetic pole are of Late Triassic to Early Jurassic age. Within the constraint that the remagnetization must be Late Triassic or younger, comparison of the tilt-corrected Atlin paleomagnetic pole with the Late Triassic to Early Jurassic North American paleomagnetic poles requires the smallest amount of interpreted tectonic motion following remagnetization. The minimum amount of latitudinal transport that this interpretation provides $-30°$. Although technically possible based on available data from the Atlin Block, the paleomagnetic data from the Takla and Hazelton groups of the Stikine terrane speak strongly against such large post-Late Triassic latitudinal transport of this region.

The abundant evidence for thermal events in the vicinity of Atlin during Cretaceous and early Tertiary time suggests that remagnetization in that time interval may, at first thought, be an attractive interpretation. However, a Cretaceous or later remagnetization is problematic. When compared with the mid-Cretaceous reference paleomagnetic pole, the *in situ* paleomagnetic pole for the Atlin Block is discordant by $\sim 40°$ and the tilt-corrected paleomagnetic pole is discordant by $\sim 70°$ (Fig. *9b).* The extreme discordance of the tilt-corrected paleomagnetic pole is ruled out by Cordilleran geology, Pacific Basin plate motion models, and all existing paleomag-

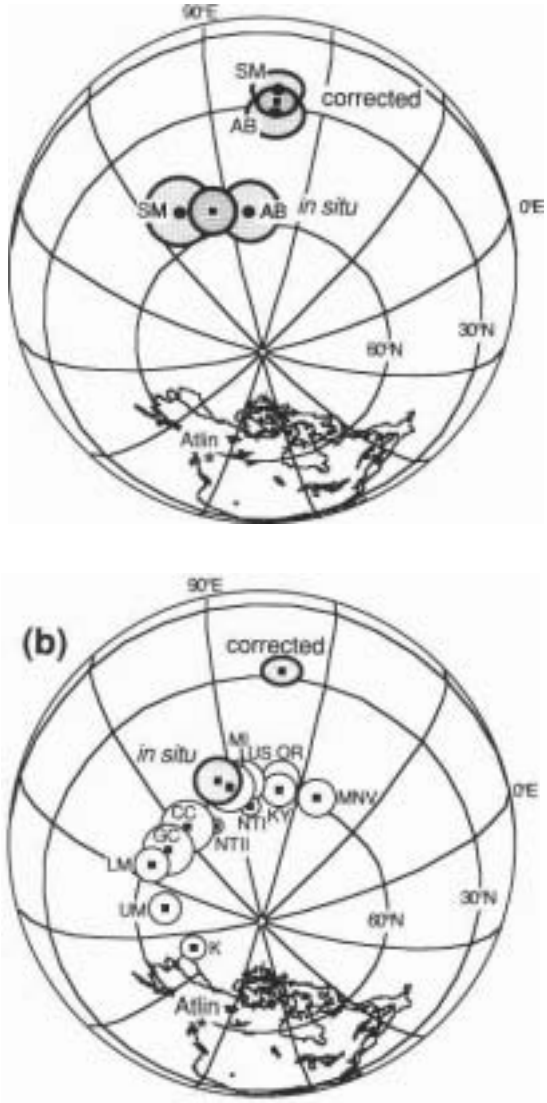


FIG. 9. (a) Paleomagnetic poles determined by averaging Alfred Butte (AB) and Sentine1 Mountain (SM) VGP's both ***in situ*** and tilt corrected. For individual collecting areas, paleomagnetic poles are shown by circles; for combined data sets, paleomagnetic poles are shown by squares. Stippled regions surrounding paleomagnetic poles are the 95% confidence limits. ***(b)*** Paleornagnetic poles determined from combined ***in situ*** and combined tilt-corrected VGP's are compared with Late Triassic through Cretaceous North American reference paleomagnetic poles. Reference poles: MI, Manicouagan impact structure (215 Ma) (Larochelle and Currie 1967; Robertson 1967); US, Upper Shale Member of Chinle Formation (−215 Ma) (Bazard and Butler 1991); MNV, Moenave Formation (−200 Ma) (Ekstrand and Butler 1989); KY, Kayenta Formation (−190 Ma) (Bazard and Butler 1991); CC, Corral Canyon (172 Ma) (May *et* al. 1986); GC, Glance Conglomerate (151 Ma) (Kluth *et* al. 1982); LM, Lower Morrison Formation (−150 Ma), UM; Upper Morrison Formation (−145 Ma) (Steiner and Helsley 1975); K, mid-Cretaceous average (Globerman and Irving 1988); NTI, Newark Trend Group I; NTII, Newark Trend Group II.

netic results from western Canada. The discordance of the ***in situ*** paleomagnetic pole when compared with a Cretaceous reference pole would require either −40° of poleward transport or −30° of southward tilting of Alfred Butte and Sentine1 Mountain since Cretaceous time. The southward tilt option would require the unreasonable scenario that the strata were steeply dipping or overturned during Cretaceous remagnetization and were subsequently modified to the more upright positions currently observed.

The interpretation of ~40° of poleward transport is also problematic. Because the Cache Creek terrane was amalgamated with the Stikine and Quesnel terranes between Late Triassic and Middle Jurassic time, any motion history implied by a Cretaceous remagnetization of the Atlin Block would also apply to the whole of Terrane I. As discussed in the Introduction, post-Late Cretaceous transport of the Atlin block is probably limited to about 1000 km by paleomagnetic data from the Carmacks Group. An interpreted Cretaceous age of remagnetization requiring 40° of latitudinal transport is quite unrealistic.

Summary

Detailed paleomagnetic analyses of samples from 33 sites at two locations within the northern portion of the Cache Creek terrane demonstrate that the majority of samples retain a well-determined ChRM. A local fold test clearly indicates a post-folding secondary origin of this ChRM that is probably the result of Late Triassic to Middle Jurassic chemical remagnetization. Lack of direct control on the age of remagnetization and on the paleohorizontal at the time of remagnetization lead to major uncertainties in tectonic interpretation of these paleomagnetic data. Perhaps acquisition of additional paleomagnetic and geologic data from elsewhere within this region will better constrain future tectonic analyses of the Cache Creek terrane.

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